Modeling Maize Response to Dynamics of Nitrogen Splits Across Growth Stages in South Africa Using DSSAT

Syferkuil seasonal analysis report based on DSSAT simulations. The objective was to evaluate how nitrogen split timing influences maize yield, biomass, nitrogen efficiency, crop growth dynamics and yield stability under historical climate variability.

1. DSSAT Simulation Overview

The analysis used four nitrogen management strategies with the same total nitrogen input of 40 kg N ha-1 but different application timings: N100 (all nitrogen at planting), N50_50 (two splits), N33_33_33 (three splits), and N25_25_25_25 (four splits). The seasonal analysis covered 36 historical seasons from 1984 to 2019, resulting in 144 treatment-season simulations.

2. Yield Response Analysis

Grain yield (HWAM) is the main indicator of maize productivity. Yield comparisons identify which nitrogen split strategy produces higher production and whether split application improves consistency across seasons.

Table 1. Grain yield performance by nitrogen treatment

Treatment	Mean_Yield	SD	Minimum	Maximum
N_100	1517.97	928.06	0	3662
N_25_25_25_25 Split	1699.31	1003.84	0	3669
N_33_33_33 Split	1671.5	1001.63	0	3672
N_50_50 Split	1504.94	892.2	0	3440

3. Biomass Production

Biomass production (CWAM) represents total crop growth. Differences among nitrogen strategies indicate how nitrogen availability influences vegetative development.

Table 2. Mean biomass production

Treatment	Mean Biomass
N_100	1517.97
N_25_25_25_25 Split	1699.31
N_33_33_33 Split	1671.5
N_50_50 Split	1504.94

4. Nitrogen Use Efficiency

Nitrogen use efficiency (NUE) was calculated as grain yield divided by total nitrogen applied. Because all treatments received 40 kg N ha-1, differences in NUE indicate the effect of nitrogen timing rather than nitrogen quantity.

Table 3. Nitrogen use efficiency

Treatment	Mean Yield	N Applied (kg N/ha)	NUE (kg grain/kg N)
N_100	1517.97	40	37.95
N_25_25_25_25 Split	1699.31	40	42.48
N_33_33_33 Split	1671.5	40	41.79
N_50_50 Split	1504.94	40	37.62

5. Crop Growth Dynamics

Plant growth analysis from plantgro.csv was used to evaluate biomass accumulation, leaf area development, grain filling and nitrogen status during the growing season. These variables explain the physiological reasons behind final yield differences.

6. Phenology Analysis

The evaluate dataset was used to compare anthesis, maturity, harvest index, biomass and yield responses among nitrogen split strategies.

7. Climate-Yield Relationship

Weather analysis evaluated rainfall, temperature and radiation variability. Rainfall-yield relationships were examined to determine whether nitrogen strategies maintained productivity under variable seasonal water availability.

8. Comparison With Lam et al. (2023)

The present study follows a similar DSSAT-based maize modelling approach but focuses specifically on nitrogen split timing. Lam et al. evaluated water and fertilizer management scenarios in South Africa, whereas this analysis investigates how the same nitrogen amount performs when applied at different crop growth stages.

9. Final Interpretation

The complete analysis combines final yield, crop growth, nitrogen uptake, efficiency, phenology and climate variability. Together these results provide evidence for selecting nitrogen application strategies that improve maize productivity and stability under South African climatic conditions.